

Quantum Field Theory I: Complete Notes

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“Nobody understands quantum mechanics.”

Richard Feynman

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1 Motivation

Quantum field theory (QFT) is the study of quantization of fields (i.e. fields made up of particles). There are three main areas of physics in which it is used:

- Critical phenomena such as phase transitions in classical statistical systems,
- Low-energy behavior of quantum many-body systems,
- Relativistic elementary particle physics (i.e. the Standard Model).

These notes are based on the class PHYS 44300 1: Quantum Field Theory I at the University of Chicago taught by Clay Cordova. As they were originally compiled in preparation for the written final exam, they are very terse and mostly focused on steps in evaluation rather than derivations. However, in assembling them in this form, I hope to provide a resource that compactly shows the relationships and intuitions behind all the formulas, which is the most difficult part of learning QFT.

2 Math background

For the remainder of the notes, we use $\int$ to denote $\int_{-\infty}^{\infty}$ by default.

2.1 Gaussian integrals

In one variable,

$$\int dx \exp\left(-\frac{1}{2}ax^2 + Jx\right) = \sqrt{\frac{2\pi}{a}} \exp\left(\frac{J^2}{2a}\right). \quad (1)$$

For a real symmetric $n \times n$ matrix A and letting J_k denote the k th column of J :

$$Z[A, J] := \int d^n x \exp\left(-\frac{1}{2}x_i A_{ij} x_j + J_i x_i\right) = \sqrt{\frac{(2\pi)^n}{\det A}} \exp\left(\frac{1}{2}J_i A_{ik}^{-1} J_k\right). \quad (2)$$

2.2 Steepest descent approximation

Let f be real and analytic on $[a, b]$ with a non-degenerate minimum at $x_0 \in [a, b]$, and $\lambda > 0$. Then:

$$\lim_{\lambda \rightarrow 0} \int_a^b dx \exp(-f(x)/\lambda) \approx \sqrt{\frac{2\pi\lambda}{f''(x_0)}} \exp(-f(x_0)/\lambda) \quad (3)$$

2.3 Feynman parameter technique

We can switch the order of p and x integration like so:

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(xA + (1-x)B)^2}. \quad (4)$$

2.4 Miscellaneous formulas

Euler product formula:

$$\sinh(\pi s) = \pi s \prod_{n=1}^{\infty} (1 + s^2/n^2).$$

Delta functions:

$$\delta^d(\vec{x}) = \int \frac{d^d k}{(2\pi)^d} e^{ikx}.$$

Gamma function:

$$\Gamma(s) = \int_0^{\infty} dx x^{s-1} e^{-x}.$$

Stirling's approximation: for $n \gg 1$,

$$n! \approx \sqrt{2\pi n} (n/e)^n.$$

Hyperbolic functions:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \cosh x = \frac{e^x + e^{-x}}{2}, \cosh^2 x - \sinh^2 x = 1.$$

Double factorial:

$$(2k)!! = 2^k k!, (2k-1)!! = \frac{(2k)!}{2^k k!}.$$

2.5 Miscellaneous physics you should already know

$$L = T - V, H = T + V = p_i \dot{x}_i - L, p_i = \frac{\partial L}{\partial \dot{x}_i}.$$

$$\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i}{\hbar} \vec{p} \cdot \vec{x}}, \vec{p} = i\hbar \frac{\partial}{\partial \vec{x}}.$$

Euler-Lagrange equations:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_i} = \frac{\partial L}{\partial x_i}.$$

Canonical equations:

$$\dot{p}_i = -\frac{\partial H}{\partial x_i}, \dot{x}_i = \frac{\partial H}{\partial p_i}.$$

Quantum harmonic oscillator:

$$E_n = \hbar\omega(n + 1/2), x = \frac{1}{\sqrt{2\omega}}(a + a^\dagger), p = -i\sqrt{\frac{\omega}{2}}(a - a^\dagger).$$

The **Lorentz group** is the group of linear transformations Λ that preserve the Lorentz metric, i.e. $\Lambda\eta\Lambda^T = \eta$. Other properties include $\det \Lambda = \pm 1, |\Lambda_0^0| \geq 1$. The Lorentz group contains four disconnected components (that is, there is a continuous transformation between any two Lorentz transformations in the same component):

1. $\Lambda_0^0 \geq 1, \det \Lambda = 1$: these are the **proper orthochronous** transformations. They preserve the sign of the time component of timelike and null vectors but not necessarily spacelike ones.
2. $\Lambda_0^0 \leq -1, \det \Lambda = 1$: these are the proper orthochronous transformations under a time reversal.
3. $\Lambda_0^0 \geq 1, \det \Lambda = -1$: these are the proper orthochronous transformations under a space reflection.
4. $\Lambda_0^0 \leq -1, \det \Lambda = -1$: these are the proper orthochronous transformations under both a time reversal and space reflection.

Perturbation theory in quantum mechanics: this often corroborates perturbative calculations with path integrals. In the time-independent case with perturbing Hamiltonian H_I ,

$$\Delta E_n^{(1)} = \langle n | H_I | n \rangle, \Delta E_n^{(2)} = - \sum_{k \neq n} \frac{|\langle k | H_I | n \rangle|^2}{E_n^{(0)}}.$$

3 Path integrals in quantum mechanics

The **path integral** in quantum mechanics allows us to exploit Lorentz-invariance of the Lagrangian in calculation of the matrix elements of the time evolution operator $U(t)$:

$$\langle x' | U(T) | x \rangle := \int_{x(0)=x}^{x(T)=x'} Dx(t) \exp \left(\frac{i}{\hbar} S[x(t)] \right). \quad (5)$$

Here $[Dx(t)]$ refers to integrating over all paths with $x(0) = x$ and $x(T) = x'$. Under $\hbar \rightarrow 0$, we just have classical physics, so for the remainder of the notes we use $\hbar = 1$.

To get convergence in Eq. (5) (i.e. damped exponents), we are motivated to evaluate the exponential of the action in a change of variables to **Euclidean/imaginary time** $\tau := it$. This gives

$$S[x(t)] = \int_0^T dt L(x(t)) = \int_0^T dt \left(\frac{1}{2} m \dot{x}^2 - V(x) \right) \implies S_E[x(t)] = \int_0^\beta d\tau (-L(x(\tau))) = \int_0^\beta d\tau \left(\frac{1}{2} m \left(\frac{dx}{d\tau} \right)^2 + V(x) \right),$$

$$Z := \langle x' | U(\beta) | x \rangle = \int_{x(0)=x}^{x(\beta)=x'} Dx(\tau) \exp(-S_E[x(\tau)]), S_E[x(\tau)] = \int_0^\beta d\tau L. \quad (6)$$

Then we can perform **Wick rotation**, which is re-evaluating Eq. (6) for imaginary τ . Note that $\tau \rightarrow \infty$ projects the system onto the ground state.

3.0.1 Harmonic oscillator

As an example evaluation, consider the harmonic oscillator

$$H = \frac{1}{2} \dot{x}^2 + \frac{1}{2} \omega^2 x^2, x(0) = x(\beta) = y. \quad (7)$$

First, find the classical solution to the imaginary time equations of motion: from the Euler-Lagrange equations,

$$L = \frac{1}{2} \dot{x}^2 - \frac{1}{2} \omega^2 x^2 \implies L_E = \frac{1}{2} \left(\frac{\partial x}{\partial \tau} \right)^2 + \frac{1}{2} \omega^2 x^2,$$

$$\implies \frac{d^2 x_0}{d\tau^2} = \omega^2 x_0 \implies x_0(\tau) = \frac{y}{\sinh(\omega\beta)} (\sinh(\omega\beta - \omega\tau) + \sinh(\omega\tau)).$$

Second, evaluate the classical action:

$$S_E[x_0(\tau)] = \int_0^\beta d\tau \left(\frac{dx}{d\tau} \right)^2 = \int_0^\beta d\tau \left(\frac{d}{d\tau} \left(x \frac{dx}{d\tau} \right) - x \frac{d^2 x}{d\tau^2} \right) = \frac{1}{2} \left(x \frac{dx}{d\tau} \right) \Big|_0^\beta + \frac{1}{2} \int_0^\beta d\tau \left(-x \frac{d^2 x}{d\tau^2} + \omega^2 x^2 \right) = \omega y^2 \tanh(\omega\beta/2).$$

Third, perform a shift of variables in the path integral:

$$x(\tau) =: x_0(\tau) + z(\tau) \implies z(0) = z(\beta) = 0, S_E[x] = S_E[x_0] + S_E[z] + \int_0^\beta d\tau \left(\frac{dx_0}{d\tau} \frac{dz}{d\tau} + \omega^2 x_0 z \right) = S_E[x_0] + S_E[z],$$

$$\Rightarrow Z = \int dy \exp(-S_E[x_0]) \int_{z(0)=0}^{z(\beta)=0} Dz \exp(-S_E[z]) = \sqrt{\frac{\pi}{\omega \tanh(\omega\beta/2)}} \int_{z(0)=0}^{z(\beta)=0} Dz \exp(-S_E[z]).$$

Now we have isolated the y -dependence of the path.

Fourth, evaluate the path integral by finding an expression for Dz :

$$z(\tau) = \sum_{\ell=1}^{\infty} c_{\ell} \sqrt{2/\beta} \sin\left(\frac{\pi\ell\tau}{\beta}\right) \Rightarrow \int Dz = \prod_{\ell} \int dc_{\ell},$$

$$\Rightarrow \int Dz \exp(-S_E[z]) = \int \prod_{\ell} dc_{\ell} \exp\left(-\frac{1}{2} \sum_{\ell} \left(\left(\frac{\pi\ell}{\beta}\right)^2 + \omega^2\right) c_{\ell}^2\right) = \mathcal{N}(\beta, \omega) := \prod_{\ell=1}^{\infty} \sqrt{\frac{2\pi\beta^2}{\pi^2\ell^2 + \omega^2\beta^2}}.$$

Fifth, perform **regularization**. The previous formal answer $\mathcal{N}(\beta, \omega)$ does not converge so we must cut off contributions from large ℓ . In our case, this can be done by dividing out the contribution from some reference frequency $\Omega \gg \omega$:

$$\frac{\mathcal{N}(\beta, \omega)}{\mathcal{N}(\beta, \Omega)} = \prod_{\ell=1}^{\infty} \sqrt{\frac{\pi^2\ell^2 + \Omega^2\beta^2}{\pi^2\ell^2 + \omega^2\beta^2}} = \sqrt{\frac{\omega \sinh(\Omega\beta)}{\Omega \sinh(\omega\beta)}} \approx \sqrt{\frac{\omega}{2 \sinh(\omega\beta)}} \exp\left(\frac{1}{2}\Omega\beta - \frac{1}{2}\ln \Omega + \dots\right).$$

Under $\Omega \rightarrow \infty$, treat the divergent terms in the above series expansion as **counterterm** contributions to the action with Lagrangian L_{ct} :

$$S_E = \int_0^{\beta} d\tau \left(\frac{1}{2} \left(\frac{dx}{d\tau} \right)^2 + \frac{1}{2} \omega^2 x^2 \right) + \int_0^{\beta} d\tau L_{ct}.$$

To get L_{ct} to converge, it must be extensive with β , depend on Ω such that $\lim_{\Omega \rightarrow \infty} Z$ exists, and contain some $\delta(\tau)$ to cancel out β -independent terms. Then (for some finite C independent of β and Ω):

$$L_{ct} = \frac{1}{2}\Omega - \frac{1}{2}\delta(\tau) \ln \Omega - C\delta(\tau) \Rightarrow \frac{\mathcal{N}(\beta, \omega)}{\mathcal{N}(\beta, \Omega)} e^{-S_{ct}} = \sqrt{\frac{\omega}{2 \sinh(\beta\omega)}} \exp\left(\frac{1}{2}\Omega\beta - \frac{1}{2}\ln \Omega\right) \exp\left(-\frac{1}{2}\Omega\beta + \frac{1}{2}\ln \Omega + C\right),$$

$$\Rightarrow \mathcal{N}(\beta, \omega) = \sqrt{\frac{A\omega}{\sinh(\beta\omega)}} \Rightarrow Z = \sqrt{\frac{A\pi}{\cosh(\beta\omega) - 1}}$$

for some constant A .

Sixth, perform **renormalization**, i.e. compare the result to the known answer. In our case,

$$\lim_{\beta \rightarrow 0} Z = \frac{e^{-\beta\omega/2}}{1 - e^{-\omega\beta}} = \sum_{n=0}^{\infty} \exp(-\beta\omega(n + 1/2)).$$

3.0.2 Interaction picture

For the Lagrangian $L = \frac{1}{2}m\dot{x}^2 - V(x) = H_0 - V(x)$, define

$$\hat{V}(t', x) = \exp(iH_0 t') V(x) \exp(-iH_0 t'). \quad (8)$$

Then

$$U(t) = \exp(-iH_0 t) T \left\{ \exp\left(-i \int_0^t dt' \hat{V}(t', x)\right) \right\}. \quad (9)$$

4 Correlation functions in quantum mechanics

The matrix element of some $A(\hat{x}(t))$ is evaluated as the **insertion of** $A(\hat{x}(t))$:

$$\langle x_f, t_f | A(\hat{x}(t)) | x_i, t_i \rangle = \int_{x(t_i)=x_i}^{x(t_f)=x_f} Dx \exp(iS[x(t)]) A(x(t)). \quad (10)$$

Thus $\int Dx \exp(iS[x])$ can be thought of as the probability measure and $A(x(t))$ as the expectation.

4.1 Time ordering

For two operators $\hat{O}_1(t_1)$ and $\hat{O}_2(t_2)$, the **time-ordered product** is

$$T\{\hat{O}_1(t_1)\hat{O}_2(t_2)\} := \Theta(t_1 - t_2)\hat{O}_1(t_1)\hat{O}_2(t_2) + \Theta(t_2 - t_1)\hat{O}_2(t_2)\hat{O}_1(t_1), \quad (11)$$

with the property

$$\int Dx \exp(iS[x]) \prod_{k=1}^n A_k(x(t_k)) = \langle x_f, t_f | T \left\{ \prod_{k=1}^n A_k(x(t_k)) \right\} | x_i, t_i \rangle. \quad (12)$$

4.2 Correlation functions

The **correlator** or **Green's function** of operators $\hat{x}(t_1), \dots, \hat{x}(t_n)$ is

$$G(t_1, \dots, t_n) := \langle x(t_1) \cdots x(t_n) \rangle := \langle 0 | \hat{x}(t_1) \cdots \hat{x}(t_n) | 0 \rangle. \quad (13)$$

When $t_1 = \dots = t_n = 0$, Eq. (13) is the **n th moment of the ground state wave function**.

In general, the **two-point function** or **contraction** is

$$G(\tau) := \overline{x(\tau)x(0)} := \langle 0 | \hat{x}(\tau)\hat{x}(0) | 0 \rangle = \lim_{\beta \rightarrow 0} \frac{\langle \psi | \exp(-(H - E_0)\beta) \hat{x}(\tau) \hat{x}(0) \exp(-(H - E_0)\beta) | \psi \rangle}{\langle \psi | \exp(-2(H - E_0)\beta) | \psi \rangle} \quad (14)$$

$$\begin{aligned} &= \frac{\int Dx \exp(-S_E[x]) x(\tau) x(0)}{\int Dz \exp(-S_E[z])} \\ &= \sum_n |\langle 0 | \hat{x} | n \rangle|^2 e^{-(E_n - E_0)\tau}. \end{aligned} \quad (15)$$

Eq. (15) is called the **Lehman spectral representation**.

When evaluating Eq. (4.2) for real time insertions, it is necessary to perform a contour integral along the imaginary axis that includes all of the real time insertions. As a result, $G(\tau)$ is automatically time-ordered.

Note that $Z[A, J] = Z[J]$ as defined in Eq. (2) is a **generating function** for correlation functions, since we can choose A to be a differential operator such that $S_E[x] = \frac{1}{2}xAx$:

$$\begin{aligned} J = (J_{k_1}, \dots, J_{k_\ell}), \langle x_{k_1} \cdots x_{k_\ell} \rangle &= \frac{1}{Z[A, 0]} \int d^n x e^{-\frac{1}{2}xAx} x_{k_1} \cdots x_{k_\ell} = \frac{1}{Z[A, 0]} \left(\frac{\partial}{\partial J_{k_1}} \cdots \frac{\partial}{\partial J_{k_\ell}} Z[A, J] \right)_{J=0} \\ &= \left(\frac{\partial}{\partial J_{k_1}} \cdots \frac{\partial}{\partial J_{k_\ell}} e^{\frac{1}{2}J^T A^{-1}J} \right)_{J=0} \end{aligned} \quad (16)$$

4.3 Wick's theorem

Wick's theorem states

$$\langle x_{a_1} x_{a_2} \cdots \rangle = \sum_{\text{all distinct contractions of pairs}} \overbrace{x_{a_1} x_{a_2} \cdots}^{\text{diagram showing pairings}} \quad (17)$$



4.3.1 Harmonic oscillator

For this section, let $m, \hbar \rightarrow 1$ and H as in Eq. (7).

$$Z[J(\tau)] = \int Dx \exp \left(-S_E[x] + \int d\tau' J(\tau') x(\tau') \right) \Rightarrow G(\tau) = \frac{1}{Z[0]} \frac{\delta}{\delta J(\tau)} \frac{\delta}{\delta J(0)} Z[J(\tau)] \big|_{J=0},$$

$$S_E = \int d\tau \left(\frac{1}{2} \dot{x}^2 + \frac{1}{2} \omega^2 x^2 \right) = \int d\tau \left(-\frac{1}{2} \ddot{x}x + \frac{1}{2} \omega^2 x^2 \right) = \frac{1}{2} \int d\tau x(\tau) \left(-\frac{d^2}{d\tau^2} + \omega^2 \right) x(\tau),$$

$$\Rightarrow A[x(\tau)] = \left(-\frac{d^2}{d\tau^2} + \omega^2 \right) x(\tau),$$

$$\Rightarrow x(\tau) = A[A^{-1}[x(\tau)]] = \int d\tau' \left(-\frac{d^2}{d\tau^2} + \omega^2 \right) G(\tau, \tau') x(\tau') \Rightarrow \left(-\frac{d^2}{d\tau^2} + \omega^2 \right) G(\tau, \tau') = \delta(\tau - \tau').$$

Then in k -space,

$$G(\tau) = \int \frac{dk}{2\pi} \tilde{G}(k) e^{ik\tau}, \delta(\tau) = \int \frac{dk}{2\pi} e^{ik\tau},$$

$$\tilde{G}(k) = \frac{1}{k^2 + \omega^2}, G(\tau) = \frac{1}{2\omega} e^{-\omega\tau}. \quad (18)$$

4.4 Analytic continuation in real time

Consider Eq. (18) derived in the previous subsection. Since this is in imaginary time, we can derive various Green's functions in real time by applying their corresponding pole prescriptions.

To get the original Green's function as in Eq. (18), force time-ordering by shifting the real-time insertions in the imaginary axis by an infinitesimal amount:

$$\tilde{G}(k) = \frac{i}{k^2 - \omega^2} = \frac{i}{k^2 - \omega^2 + i\epsilon} \Rightarrow G(t) = \frac{1}{2\omega} e^{-i\omega t}. \quad (19)$$

Equivalently, time-ordering is forced because we can choose the contour along which to perform the inverse Fourier transform. If $t > 0$, we select for the pole in the positive imaginary axis. If $t < 0$, we select for the pole in the negative imaginary axis.

To get the **retarded Green's function**, we move both poles forward in imaginary time:

$$\tilde{G}_R(k) = \frac{i}{(k - i\epsilon)^2 - \omega^2} \Rightarrow G_R(t) = \Theta(t) \langle 0 | [x(t), x(0)] | 0 \rangle = \frac{1}{2\omega} (e^{-i\omega t} - e^{i\omega t}). \quad (20)$$

To get the **advanced Green's function**, we move both poles backward in imaginary time:

$$\tilde{G}_A(k) = \frac{i}{(k + i\epsilon)^2 - \omega^2}, \quad (21)$$

4.5 Feynman diagrams

4.5.1 Anharmonic oscillator

Consider a harmonic oscillator with a small perturbation:

$$H = \frac{p^2}{2} + \frac{1}{2}\omega^2 x^2 + \frac{\lambda}{4!}x^4.$$

Then

$$\begin{aligned} Z_\lambda &:= \int Dx e^{-S_E[x]} = \int Dx \exp\left(-\int d\tau \left(\frac{\dot{x}^2}{2} + \frac{1}{2}\omega^2 x^2\right)\right) \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{4!} \int d\tau' x^4(\tau')\right)^n \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{4!}\right)^n \int d\tau_1 \cdots d\tau_n \left(\int Dx \exp\left(-\int d\tau \left(\frac{\dot{x}^2}{2} + \frac{1}{2}\omega^2 x^2\right)\right) x^4(\tau_1) \cdots x^4(\tau_n)\right) \\ &= Z_0 \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{4!}\right)^n \int d\tau_1 \cdots d\tau_n \langle x^4(\tau_1) \cdots x^4(\tau_n) \rangle_{\lambda=0}. \end{aligned} \quad (22)$$

Thus, we can compute Z_λ to the n th order using Wick's theorem with contractions of the original Hamiltonian. More specifically, these and general correlation functions such as the **two-point function**

$$G_\lambda(\tau) := \frac{\int D(x) e^{-S_E[x]} x(\tau) x(0)}{Z_\lambda} \quad (23)$$

are computed using **Feynman diagrams**. In general, this should converge as an asymptotic series in n (even though the total sum diverges).

4.5.2 In position space

For calculation of Z_λ/Z_0 to the n th order with a perturbation of form $\frac{\lambda}{a!}x^a$:

- Feynman diagram contains n vertices with valence a each representing some $x(\tau_i)$.
- For diagram Γ , each internal vertex contributes $-\lambda \int d\tau_i$ and each internal edge contributes $G_0(\tau_1 - \tau_2)$, where $G_0(\tau)$ is the unperturbed theory contraction as in Eq. (18).
- Total contribution of diagram is $\Gamma/S(\Gamma)$, where $S(\Gamma)$ is the **symmetry factor** that counts the number of automorphisms of the graph.

$$\# \text{ of Wick terms} = \frac{(a!)^n n!}{S(\Gamma)}. \quad (24)$$

A diagram is **connected** if each vertex is connected to another. Define $Z^{(n)}$ as the sum over all connected diagrams of order n . Then

$$Z_\lambda/Z_0 = \exp\left(\sum_{n=1}^{\infty} Z^{(n)}\right) = \lim_{T \rightarrow \infty} \exp(-T \Delta E_0) \implies \Delta E_0(\lambda) = \lim_{T \rightarrow \infty} -\frac{1}{T} \sum_{n=1}^{\infty} Z^{(n)}. \quad (25)$$

For calculation of

$$G_\lambda(\tau_1 - \tau_2)(Z_\lambda/Z_0) = \frac{\int Dx e^{-S_E[x]} x(\tau_1) x(\tau_2)}{Z_0} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{\lambda}{a!}\right)^n \int d\tau'_1 \cdots d\tau'_n \langle x(\tau_1) x(\tau_2) x^a(\tau'_1) \cdots x^a(\tau'_n) \rangle_{\lambda=0}$$

the rules are the same as above but with the addition of two non-interchangeable external points representing $x(\tau_1)$ and $x(\tau_2)$.

A diagram is **1-particle irreducible (1PI)** if it remains connected after cutting one internal line. Define the **amputated diagram** or **self-energy** in momentum space as

$$\sum(k) := \sum -^k(\text{1PI diagram}) -^k \quad (26)$$

which does not include G_0 from the external points. Then

$$\tilde{G}_\lambda(k) = \frac{1}{k^2 + \omega^2 - \sum(k)}, G_\lambda(\tau) = \int \frac{dk}{2\pi} e^{ik\tau} \tilde{G}_\lambda(k). \quad (27)$$

4.5.3 In momentum space

For calculation of $\tilde{G}_\lambda(k)$ to the n th order for a perturbation of form $\frac{\lambda}{a!} x^a$:

- Same number of internal and external vertices + valence as previous.
- Each contraction/line contributes $\tilde{G}_0(k)$ and has a directional flow of k . Each internal vertex represents momentum conservation given by $-\lambda(2\pi)\delta(k_1 + \dots + k_a)$.
- For I internal lines and V vertices there are $I - V + 1$ loops contributing $\int \frac{d\ell}{2\pi}$ (i.e. integration over unbounded momentum).
- Total contribution of diagram is $\Gamma/S(\Gamma)$.

Note that for calculation of Z_λ/Z_0 in momentum space (i.e. same rules but without external points), we just end up with the coefficient of T , as in Eq. (25).

5 Second quantization

In quantum mechanics, we get a violation of causality due to unbounded speed of the free particle. This motivates us to look at field theory. As a review,

$$S = \int dt L, L = \int d^{d-1}x \mathcal{L}, \pi = \frac{\partial \mathcal{L}}{\partial_t \varphi}, H = \int d^{d-1}x \mathcal{H}, \mathcal{H} = \pi \partial_t \varphi - \mathcal{L},$$

and the Euler-Lagrange equations become

$$\frac{\partial \mathcal{L}}{\partial \varphi_a} = \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\nu \varphi_a)} \right).$$

There are two ways to think about quantizing fields- the first is canonical quantization, which is an extension of the Poisson bracket in classical field theory to the equal-time commutator in quantum mechanics:

$$\{\varphi(\vec{x}), \pi(\vec{y})\} = \delta^{d-1}(\vec{x} - \vec{y}) \implies [\phi(\vec{x}), \pi(\vec{y})] = i\delta^{d-1}(\vec{x} - \vec{y}). \quad (28)$$

This can be thought of as a collection of infinite quantum harmonic oscillators at point x in spacetime. The second is path integral quantization, which we discuss in Section 6.

5.1 Real scalar field

The **real scalar field** is

$$\phi(x) = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \frac{1}{\sqrt{2\omega_p}} (a_{\vec{p}} e^{ip_\nu x^\nu} + a_{\vec{p}}^\dagger e^{-ip_\nu x^\nu}) = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \frac{1}{\sqrt{2\omega_p}} (a_{\vec{p}} + a_{-\vec{p}}^\dagger) e^{ip_\nu x^\nu}, p_0 := \omega_p := \sqrt{\vec{p}^2 + m^2} \quad (29)$$

with free/unperturbed Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\nu\phi\partial^\nu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}(\partial_0\phi)^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2, \quad (30)$$

mode algebra

$$[a_{\vec{p}}, a_{\vec{q}}^\dagger] = (2\pi)^{d-1}\delta^{d-1}(\vec{p} - \vec{q}), \quad (31)$$

and Hamiltonian

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}|\nabla\phi|^2 + \frac{1}{2}m^2\phi^2 \implies H = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \omega_p a_{\vec{p}}^\dagger a_{\vec{p}}. \quad (32)$$

Note that this creates a Hilbert space of bosons called the **Fock space** given by

$$\bigoplus_{\ell=0}^{\infty} \{(a_{\vec{p}_1}^\dagger)^{n_1} \dots (a_{\vec{p}_n}^\dagger)^{n_k} |0\rangle \mid n_1 + \dots + n_k = \ell\} \quad (33)$$

where each such vector has energy $E = n_1\omega_{p_1} + \dots + n_k\omega_{p_k}$. To force Lorentz-invariance, we also define

$$|\vec{p}\rangle = \sqrt{2\omega_p} a_{\vec{p}}^\dagger |0\rangle \implies \langle \vec{p} | \vec{q} \rangle = 2\omega_p (2\pi)^{d-1} \delta^{d-1}(\vec{p} - \vec{q}), \quad (34)$$

Note that the real scalar Lagrangian density has a symmetry under $\phi_a \mapsto T_{ab}\phi_b$, where T_{ab} is a transformation in $SO(3)$ with $T = \exp(-\sum_\ell \alpha_\ell L_\ell)$, where L_ℓ are the 3D rotation generators.

5.2 Causality

In a general scalar field theory, the Lorentz-invariant QFT analog to $\langle \vec{y}, t \mid \vec{x}, 0 \rangle$ is the **propagator**

$$D(y-x) := \langle 0 \mid \phi(y)\phi(x) \mid 0 \rangle = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \frac{1}{2\omega_p} e^{ip_\nu(y-x)^\nu}, \tilde{D}(k) = 2\pi\Theta(k^0)\delta(k_\nu k^\nu + m^2). \quad (35)$$

Note that if x is timelike, $D(x) \neq 0$. However if x is spacelike, $\lim_{|\vec{x}| \rightarrow \infty} D(x) \neq 0$. This does not necessarily violate causality under certain conditions.

For example, suppose we have a localized perturbation

$$\delta H(t) \sim \int d^{d-1}y \hat{A}(t, \vec{y}) J(t, \vec{y})$$

for some localized operator $\hat{A}(t, \vec{y})$ and source function J . Then the response of the measurement $\hat{B}(t, \vec{x})$ in $|\psi\rangle$ is given by the **Kubo formula**:

$$\begin{aligned} \delta(\langle \psi \mid \hat{B}(x) \mid \psi \rangle) &= -i \int d^d y \Theta(x^0 - y^0) J(y) \langle \psi \mid [\hat{B}(x), \hat{A}(y)] \mid \psi \rangle, \\ J(z) = \delta^d(z - y) &\implies \delta(\langle \psi \mid \hat{B}(x) \mid \psi \rangle) = -i\Theta(x^0 - y^0) \langle \psi \mid [\hat{B}(x), \hat{A}(y)] \mid \psi \rangle. \end{aligned} \quad (36)$$

Thus, the **causality criterion in QFT** (i.e. **microcausality**) is:

$$\begin{aligned} |\psi\rangle \text{ has signal at } x \text{ that vanishes when spacelike-separated from a perturbation at } y \\ \iff \text{For any localized operators } \hat{A} \text{ and } \hat{B}, [\hat{A}(x), \hat{B}(y)] = 0 \text{ for spacelike } x - y. \end{aligned}$$

The equal time commutation relations in Eq. (28) are compatible with causality. The general time commutator is

$$[\phi(x), \phi(y)] = D(x-y) - D(y-x). \quad (37)$$

Then we define the **retarded propagator** as

$$D_R(z) := \Theta(z^0) \langle 0 \mid [\phi(z), \phi(0)] \mid 0 \rangle = \Theta(z^0)(D(z) - D(-z)), \tilde{D}_R(p) = \frac{i}{(p^0 + i\epsilon)^2 - \vec{p}^2 - m^2}. \quad (38)$$

5.3 Complex scalar field

The **complex scalar field** is the smallest model with charge conservation given by

$$\phi(x) = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \frac{1}{\sqrt{2\omega_p}} (a_{\vec{p}} e^{-ip_\nu x^\nu} + b_{\vec{p}}^\dagger e^{ip_\nu x^\nu}) \quad (39)$$

with free/unperturbed Lagrangian density

$$\mathcal{L} = -\partial_\nu \phi \partial^\nu \phi^* - m^2 \phi \phi^*, \quad (40)$$

mode algebra

$$[a_{\vec{p}}, a_{\vec{q}}^\dagger] = [b_{\vec{p}}, b_{\vec{q}}^\dagger] = (2\pi)^{d-1} \delta^{d-1}(\vec{p} - \vec{q}) \implies [\phi(\vec{x}), \pi(\vec{y})] = [\phi^*(\vec{x}), \pi^*(\vec{y})] = i\delta^{d-1}(\vec{x} - \vec{y}), \quad (41)$$

and Hamiltonian

$$\mathcal{H} = \pi \dot{\phi} + \pi^* \dot{\phi}^* - \mathcal{L} \implies H = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} \omega_p (a_{\vec{p}}^\dagger a_{\vec{p}} + b_{\vec{p}}^\dagger b_{\vec{p}}). \quad (42)$$

The conserved Noether current (for $\phi \mapsto e^{i\alpha} \phi$) is

$$J^\nu = i\phi^* \partial^\nu \phi - i\phi \partial^\nu \phi^*,$$

which gives a Lorentz-invariant conserved **charge**

$$Q = \int d^{d-1}x J^0 = \int \frac{d^{d-1}p}{(2\pi)^{d-1}} (a_{\vec{p}}^\dagger a_{\vec{p}} - b_{\vec{p}}^\dagger b_{\vec{p}}). \quad (43)$$

Note that the $b_{\vec{p}}, b_{\vec{p}}^\dagger$ modes correspond to **antiparticles** with the same mass/energy as the $a_{\vec{p}}, a_{\vec{p}}^\dagger$ modes and opposite charge.

Note that the complex scalar Lagrangian density has a symmetry under $\phi_a \mapsto T_{ab} \phi_b$, where T_{ab} is a transformation in $SU(2)$ with $T = \exp(-\sum_\ell \alpha_\ell \sigma_\ell)$, where L_ℓ are the Pauli matrices. $SU(2)$ and $SO(3)$ are isomorphic in this way.

5.4 Noether's theorem

For a Lagrangian density made up of fields ϕ_a , suppose that under the transformation

$$x^\mu \mapsto x'^\mu := x^\mu + \epsilon \delta x^\mu, \phi_a \mapsto \phi'_a := \phi_a + \epsilon \delta \phi_a,$$

we have

$$\int d^D x' \mathcal{L}' = \int d^D x (\mathcal{L} + \partial_\mu Y^\mu).$$

Then the **conserved Noether current** is

$$J^\mu := \delta x^\mu \mathcal{L} - \delta x^\nu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\nu \phi_a + \delta \phi_a \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} - Y^\mu, \partial_\mu J^\mu = 0. \quad (44)$$

In particular, the **energy-momentum tensor** of the real scalar field is derived using the transformation

$$\begin{aligned} x^\mu \mapsto x^\mu + a^\mu &\implies \phi \mapsto \phi + a_\mu \delta \phi_\mu, \mathcal{L} \mapsto \mathcal{L} + a_\mu \partial_\mu \Lambda^\mu, \\ \implies T^\mu{}_\nu &:= \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi_\nu - \Lambda^\mu{}_\nu = -\partial^\mu \phi \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}. \end{aligned} \quad (45)$$

This has the property

$$\int d^{d-1}x T_{00} = \int d^{d-1}x \mathcal{H} = H, \int d^{d-1}x T_{0i} = P_i := \int \frac{d^{d-1}p}{(2\pi)^{d-1}} p_i a_{\vec{p}}^\dagger a_{\vec{p}}.$$

6 Path integral quantization of scalar fields

The QFT analog to the QM path integral in Eq. (5) is

$$\langle \phi_f(\hat{x}) | e^{-iHT} | \phi_i(\hat{x}) \rangle = \int_{\phi(0)=\phi_i}^{\phi(T)=\phi_f} D\phi \exp \left(i \int_0^T dx^0 \int d^{d-1}x \mathcal{L} \right), \quad (46)$$

The QFT analog to the QM correlation function in Eq. (13) is

$$\begin{aligned} \langle 0 | T \{ \hat{\phi}(x_1) \cdots \hat{\phi}(x_n) \} | 0 \rangle &= \lim_{T \rightarrow \infty(1-i\epsilon)} \frac{\int D\phi \exp \left(i \int_{-T}^T dt L \right) \phi(x_1) \cdots \phi(x_n)}{\int D\phi \exp \left(i \int_{-T}^T dt L \right)} \\ &= \langle 0 | \hat{\phi}(x_1^E) \cdots \hat{\phi}(x_n^E) | 0 \rangle = \frac{\int D\phi e^{-S_E} \phi(x_1^E) \cdots \phi(x_n^E)}{\int D\phi e^{-S_E}}. \end{aligned} \quad (47)$$

6.0.1 Normal ordering

Since we only care about measuring energy differences (i.e. vacuum expectation values should go to 0), it is useful to define the **normal-ordered product**

$$: \phi^2(x) :: \lim_{y \rightarrow x} (\phi(x)\phi(y) - \overline{\phi(x)\phi(y)}). \quad (48)$$

This is equivalent to switching the order of mode operators in $\phi^2(x)$ so that annihilation operators are always before creation operators. Notice that normal-ordered products are not necessarily positive definite and

$$\langle 0 | : \phi^2(x) :: \phi^2(y) : | 0 \rangle = 2D^2(x - y).$$

6.1 Free scalar field correlators

The free (real) scalar Lagrangian in Euclidean time is

$$\mathcal{L} = \frac{1}{2} \partial_\nu \phi \partial^\nu \phi + \frac{1}{2} m^2 \phi^2.$$

As in Eq. (18), we derive the **Feynman propagator** (i.e. the two-point function) as the inverse of a differential operator:

$$\overline{\phi(x)\phi(y)} = G_E(x - y)_0 := \int \frac{d^d k}{(2\pi)^d} \frac{e^{ik_\nu(x-y)^\nu}}{|k|^2 + m^2}, \tilde{G}_E(k)_0 = \frac{1}{|k|^2 + m^2}, \quad (49)$$

$$\implies D_F(x - y) := \langle 0 | T \{ \phi(x)\phi(y) \} | 0 \rangle = -i \int \frac{dk^0 d^{d-1}k}{(2\pi)^d} \frac{e^{ik_\nu(x-y)^\nu}}{k_\nu k^\nu + m^2 - i\epsilon}. \quad (50)$$

Note that these are equivalent to the time-ordered version of the propagator defined in Eq. (35).

6.2 Feynman diagrams

The shapes of the Feynman diagrams in field theory are generally the same as those described in quantum mechanics (Section 4.5) but have different contributions.

6.2.1 In position space

For calculation of $(\int D\phi e^{-S_E} \phi(x_1) \cdots \phi(x_n))/Z_0$ to the n th order with a perturbation of form $\frac{\lambda}{a!} \phi^a$:

- Each vertex corresponds to a spacetime point x and contributes $-\lambda \int d^d x$. Each internal edge contributes $G_E(x-y)_0$.
- Each insertion corresponds to a non-interchangeable external point representing x_i . Each acceptable graph has every connected component connected to at least one external point.
- The total contribution of the graph is $\Gamma/S(\Gamma)$.

6.2.2 In momentum space

For calculation of $\int D\phi e^{-S_E} \phi(x_1) \cdots \phi(x_n)$ to the n th order with a perturbation of form $\frac{\lambda}{a!} \phi^a$:

- Each internal vertex represents momentum conservation given by $-\lambda(2\pi)\delta(k_1 + \cdots + k_a)$. Each edge contributes $\tilde{G}_E(k)_0 = \frac{1}{|k|^2 + m^2}$.
- Each insertion corresponds to a non-interchangeable external point with $\sum k_{external} = 0$. Each acceptable graph has every connected component connected to at least one external point.
- Integrate over unbounded momenta $\int \frac{d^d \ell}{(2\pi)^d}$ (i.e. loops- same as before).
- The total contribution of the graph is $\Gamma/S(\Gamma)$.

Note that in the calculation of amputated diagrams $\sum(k)$ we must not include any factors from the external legs.

6.3 Spectral decomposition

In real time, the QFT analog to the QM spectral representation in Eq. (15) is given by the **Wightman function**:

$$\omega(x) := \langle 0 | \phi(x) \phi(0) | 0 \rangle = \int d\beta e^{i(p_\beta)_\nu x^\nu} |\langle \beta | \phi(0) | 0 \rangle|^2 = \int d^d x e^{ik_\nu x^\nu} \int d\beta \delta^d(k - p_\beta) |\langle \beta | \phi(0) | 0 \rangle|^2, \quad (51)$$

where $\hat{p}|\beta\rangle =: p_\beta|\beta\rangle$. Then the **spectral function** can be defined as

$$\int d\beta \delta^d(k - p_\beta) |\langle \beta | \phi(0) | 0 \rangle|^2 =: \frac{\Theta(k^0)}{(2\pi)^{d-1}} \rho(-k^2 = -k_\nu k^\nu). \quad (52)$$

Note that all states have $y^2 := -k_\nu k^\nu \geq 0$, so we can write the **general correlation function**

$$\omega(x) = \int_0^\infty dy^2 \rho(y^2) D(x, y^2), \quad (53)$$

where $D(x, y^2)$ is defined based on the propagator in Eq. (35):

$$D(x, y^2) := \langle 0 | \phi(x) \phi(0) | 0 \rangle = \int \frac{d^{d-1} k}{(2\pi)^{d-1}} \frac{e^{ik_\nu x^\nu}}{2\omega_k}, \omega_k := \sqrt{\vec{k}^2 + y^2},$$

$$\tilde{D}(k, y^2) = 2\pi \Theta(k^0) \delta(k^2 + y^2). \quad (54)$$

In the free theory, $\rho(y^2) = \delta(y^2 - m^2)$.

Similarly, we can define a time-ordered version of Eq. (54) based on the Feynman propagator in Eq. (50)

$$D_F(x, m^2) := -i \int \frac{d^d k}{(2\pi)^d} \frac{e^{ik_\nu x^\nu}}{k_\nu k^\nu + m^2 - i\epsilon} = \langle 0 | T\{\phi(x) \phi(0)\} | 0 \rangle \quad (55)$$

with corresponding **time-ordered correlator**

$$\langle 0 | T\{\phi(x) \phi(0)\} | 0 \rangle = \int_0^\infty dy^2 \rho(y^2) D_F(x, y^2). \quad (56)$$

6.3.1 Kallen-Lehmann spectral representation

Suppose we have a theory with a single species of particle with mass m . Then either

- $|\beta\rangle = |0\rangle \implies y = 0$,
- $|\beta\rangle$ is a one-particle state $\implies y = m$,
- $|\beta\rangle$ is a multiparticle state. Since we work in the center of mass frame, we have $y^2 = -p_\nu p^\nu = 2\sqrt{\vec{p}^2 + m^2} > 4m^2$.

Then, the **Kallen-Lehmann spectral representation** splits the spectral function into the single-particle (given by the delta function) and multiparticle (given by some $\sigma(y^2)$ which has support at $y \geq 2m$) contributions:

$$\rho(y^2) = Z\delta(y^2 - m^2) + \sigma(y^2) \implies \omega(x) = ZD(x, m^2) + \int_{4m^2}^{\infty} dy^2 \sigma(y^2) D(x, y^2), \quad (57)$$

where Z is the **field renormalization**.

6.4 Renormalization

In an interacting theory, we can determine the spectral function in Eq. (52) by first evaluating the Euclidean-time correlator $\tilde{G}_E(k)_\lambda = \tilde{G}_E(k)$ and then applying the **Stieltjes transform**

$$\tilde{G}_E(k) = \int_0^\infty dy^2 \frac{1}{|k|^2 + y^2} \rho(y^2) = \frac{1}{|k|^2 + m^2 - \sum(k)}. \quad (58)$$

6.4.1 ϕ^4 theory

Consider a perturbation of the free real scalar Lagrangian density by $\frac{\lambda}{4!}\phi^4$. To the first order in k -space,

$$\sum(k) = -\frac{\lambda}{2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{|k|^2 + m^2} = -\frac{\lambda}{2} \frac{\text{Vol}(d-1)}{(2\pi)^d} \int_0^\infty dk \frac{k^{d-1}}{k^2 + m^2} \sim \int_0^\infty dk k^{d-3}.$$

This integral diverges when $d \geq 2$. To deal with this, we first apply **regularization** (i.e. modifying the path integral measure to get a finite result). This can be done by applying a **finite momentum cutoff** $\Omega \gg m$:

$$\begin{aligned} \int D\phi &= \int \prod_k d\phi(k) \mapsto \int \prod_{k \leq \Omega} d\phi(k), \quad \sum(k) =: -\lambda S(\Omega), \\ \implies S(\Omega) &:= \int_0^\Omega dk \frac{k^{d-1}}{k^2 + m^2} \approx \int_0^\Omega dk k^{d-3} \sum(-m^2/k^2)^n = \begin{cases} \frac{1}{2} \ln(1 + \Omega^2/m^2) & d = 2 \\ \Omega - m \tan^{-1}(\Omega/m) & d = 3 \\ \frac{1}{2} \left(\Omega^2 - m^2 \ln \left(\frac{m^2 + \Omega^2}{m^2} \right) \right) & d = 4. \end{cases} \end{aligned}$$

Note that there are two masses: the **bare mass** m which is the parameter in the Lagrangian density, and the *physical mass of the particle* m_p - this is the mass we refer to in the KL-spectral representation in Eq. (57). Then we can write

$$m^2 - \sum(k) = m_p^2$$

and define δm^2 as the **mass counterterm** such that

$$m^2 =: m_R^2 + \delta m^2, \delta m^2 = \sum_{n=1}^{\infty} \lambda^n (\delta m^2)^{(n)} \implies m_p^2 = m_R^2 + (\delta m^2)^{(1)} + \lambda S(\Omega) \text{ converges under } \Omega \rightarrow \infty.$$

We can view such counterterms as part of the **counterterm Lagrangian** $\delta\mathcal{L}$, i.e. continuing to work in Euclidean time,

$$\mathcal{L} = \mathcal{L}_R + \delta\mathcal{L}, \frac{1}{2}m_R^2\phi^2 \subset \mathcal{L}_R, \frac{1}{2}\delta m^2\phi^2 \subset \delta\mathcal{L}.$$

6.4.2 ϕ^3 theory

Consider a perturbation of the free real scalar Lagrangian density by $\frac{\lambda}{3!}\phi^3$. To the second order in k -space,

$$\sum(k) = \frac{\lambda^2}{2} \int \frac{d^d p}{(2\pi)^d} \frac{1}{|p|^2 + m^2} \frac{1}{|k+p|^2 + m^2} = \frac{\lambda^2}{2} \frac{\text{Vol}(S^{d-1})}{(2\pi)^d} \int_0^1 dx \int_0^\infty d\ell \frac{\ell^{d-1}}{(\ell^2 + x(1-x)k^2 + m^2)^2}.$$

The regularized self-energy is then

$$\begin{aligned} \sum_\Omega(k) &= \frac{\lambda^2}{2} \frac{\text{Vol}(S^{d-1})}{(2\pi)^d} \int_0^1 dx \int_0^\Omega d\ell \frac{\ell^{d-1}}{(\ell^2 + x(1-x)k^2 + m^2)^2} \\ &= \frac{1}{2} \frac{\text{Vol}(S^{d-1})}{(2\pi)^d} \int_0^1 dx \int_0^\Omega d\ell \sum_{n=0}^\infty (-1)^n (n+1) \Delta^n \ell^{d-5-2n}, \Delta := x(1-x)k^2 + m^2. \end{aligned}$$

For any d , this contains finitely many divergent terms, so we need finitely many corresponding counterterms. As an example, consider $d = 6$:

$$\sum_\Omega(k^2) = \frac{\lambda^2}{256\pi^3} \int_0^1 dx \left(\Omega^2 - 2\Delta \ln \left(\frac{\Omega^2 + \Delta}{\Delta} + \Delta - \frac{\Delta^2}{\Omega^2 + \Delta} \right) \right) \Rightarrow \delta L_E = \frac{1}{2} \delta m^2 \phi^2 + \frac{1}{2} \delta Z \partial_\mu \phi \partial^\mu \phi,$$

where δZ is the **wavefunction renormalization**. Note that the mass counterterm absorbs up to k -dependent divergences and the wavefunction renormalization absorbs k^2 -dependent divergences. We define δZ , δm^2 such that

$$\sum(k) = \sum_\Omega(k^2) - \delta m^2 - k^2 \delta Z$$

converges and

$$\tilde{G}_E(k^2) = \frac{1}{k^2 + m_p^2 - \sum(k^2)}$$

as in Eq. (58) has a pole at $k^2 = -m_p^2$ with residue 1. Then

$$\begin{aligned} \delta Z &= \sum'_\Omega(-m_p^2), \delta m^2 = \sum_\Omega(m_p^2) + m_p^2 \sum'_\Omega(-m_p^2), \\ \Rightarrow \sum(k^2) &= \frac{\lambda^2}{128\pi^3} \int_0^1 dx (\Delta \ln(\Delta/\Delta_*) - (\Delta - \Delta_*)), \Delta_* := \Delta|_{k^2=-m_p^2}. \end{aligned}$$

Note that $\tilde{G}_E(-z) := \tilde{G}_E(k^2)$ (with $s = y^2$) is analytic away from $z \geq 0$, so we expect a *branch cut* at $z \geq 0$. That is, there is a discontinuity

$$\frac{1}{2\pi i} (\tilde{G}_E(-z - i\epsilon) - \tilde{G}_E(-z + i\epsilon)) = \frac{1}{2\pi i} \int_0^\infty ds \rho(s) \left(\frac{1}{s - z - i\epsilon} - \frac{1}{s - z + i\epsilon} \right) = \int_0^\infty ds \rho(s) \frac{1}{\pi} \frac{\epsilon}{(s - z)^2 + \epsilon}.$$

Taking the $\epsilon \rightarrow 0$ limit allows us to evaluate the spectral function as

$$\rho(s) = \delta(s - m_p^2) + \frac{\lambda^2}{768\pi^3} \Theta(s - 4m_p^2) \frac{(1 - 4m_p^2/s^2)^{3/2}}{(s - m_p^2)^2}.$$

6.5 Scattering theory

In a collision event, the “in” states $\{|\vec{k}_1, \dots, \vec{k}_m\rangle^{in}\}$ and “out” states $\{|\vec{p}_1, \dots, \vec{p}_n\rangle^{out}\}$ form different bases for the same Hilbert space. Then we define the **S-matrix** as

$$S_{\vec{p}_1, \dots, \vec{p}_n; \vec{k}_1, \dots, \vec{k}_m} := {}^{out} \langle \vec{p}_1, \dots, \vec{p}_n | \vec{k}_1, \dots, \vec{k}_m \rangle^{in}, |\alpha\rangle^{in} = \int d\beta S_{\beta\alpha} |\beta\rangle^{out}. \quad (59)$$

Whereas in statistical physics, correlation functions are the fundamental observables, S-matrices are the fundamental observables in particle physics.

By the **LSZ reduction formula**, under the equal pole limit $p_i^0 \rightarrow E_{p_i} := \sqrt{\vec{p}_i^2 + m^2}, k_j^0 \rightarrow E_{k_j} = \sqrt{\vec{k}_j^2 + m^2}$:

$$\begin{aligned} & \int \prod_{i=1}^n d^d x_i e^{-ip_i x_i} \prod_{j=1}^m d^d y_j e^{ik_j y_j} \langle 0 | T \{ \phi(x_1) \cdots \phi(x_n) \phi(y_1) \cdots \phi(y_m) \} | 0 \rangle \\ & \sim \prod_{i=1}^n \frac{i\sqrt{Z}}{-p_i^2 - m^2 + i\epsilon} \prod_{j=1}^m \frac{i\sqrt{Z}}{-k_j^2 - m^2 + i\epsilon} S_{\vec{p}_1, \dots, \vec{p}_n; \vec{k}_1, \dots, \vec{k}_m}. \end{aligned} \quad (60)$$